Privacy-Aware Continual Knowledge Distillation for Cross-Domain Medical Image Classification

Bishozit Chandra Das
Student ID: VR544246
University of Verona
bishozit.chandradas@studenti.univr.it

Abdullah Al Noman Taki
Student ID: VR528988
University of Verona
abdullahalnoman.taki@studenti.univr.it

Shaikh Faisal Rahman
Student ID: VR546093
University of Verona
shaikhfaisal.rahman@studenti.univr.it

Abstract

The work of hospitals does not cease when a model is deployed; there are new machines that scan patients and new questions that require diagnostic, and thus a model trained the previous year will have to handle information from this year. But retraining on all encountered data is impossible because storing and replaying old scans of patients is not an option due to privacy and technical limitations. Without using the old data, training a neural network on a new task results in overwriting old knowledge. We create an architecture that enables the compact network to learn new medical imaging tasks while retaining the previously learned tasks, along with the new addition of the aspect of the amount of information that is leaked by the trained model regarding its training data. Firstly, the teacher is a ResNet-50 (or ViT-B/16) network that learns chest X-ray pneumonia task (Task A), is distilled to the compact multi-head MobileNetV3 (or EfficientNet-B0) student, and further makes the student learn brain-tumour MRI task (Task B) without revisiting Task A. We analyze naive fine-tuning, EWC, LwF, and BatchNorm freezing techniques followed by a membership-inference attack and defense using DP-SGD. The distilled model achieves an accuracy of 97.0% on task A which is only slightly better than its own teacher while naive adaptation reduces task A retention to as little as 28%. The freezing of BatchNorm statistics enables almost 100% retention for task A without any effect on task B. Lastly, it was observed from the privacy analysis that the leakage from this scenario is already low (attack AUC ≈ 0.53). Code: github.com/BishozitChandraDas/Privacy_Aware_Continual_KD.

1. Introduction

Manual interpretation of images in the field of medicine takes long and the number of specialists is always insufficient. With the constantly growing number of scans, automation is becoming more and more necessary rather than an option. There are two imaging techniques frequently used in this project – chest X-rays and brain MRI, covering all kinds of diseases from pneumonia to brain tumors. The problem with manual image interpretation lies not only in the fact that it requires time but also because there is no consensus among radiologists and the same image can have two interpretations. The software that highlights critical cases and does the job of routine scans is what frees up human experts for the more important tasks. The same approach can be found in many other areas such as self-driving cars, quality control in manufacturing and security systems.

An actual clinical model will always be incomplete since there will always be new jobs for it to learn and new imaging machines coming out, meaning it has to learn them too. Re-training on all the scans it has learned up till now is impossible most of the time, not only due to lack of storage space but more importantly, because past patient data cannot simply be saved and used. Thus, the model will update itself on the new job only, which is precisely the point where it begins to forget the old one.

Various algorithmic methods have been employed to tackle the problem of forgetting. Elastic Weight Consolidation (EWC) restricts parameter update of parameters which are essential for the previous task by using Fisher-information regularization, while Learning without Forgetting (LwF) maintains previous behavior by distillation loss on a frozen model. Simultaneously, knowledge distillation transfers knowledge from a powerful teacher network into an efficient student network. However, most of the research works so far have concentrated on the algorithmic aspect while neglecting the architectural aspects; notably, Batch Normalization maintains statistics that are extremely sensitive to the change in distribution and adversely affect the previous task.

More importantly, even the very assumption of continuous learning—that older data is never kept—serves the purpose of privacy protection, but a trained model can reveal who was part of the training set anyway. Hence, we take privacy into account right from the beginning. Our contribution is: (i) A combined KD+CL framework for cross-modal medical imaging, specifically in the X-ray and MRI modalities; (ii) A comparative study on different CL approaches for naive fine-tuning, EWC, LwF, and BatchNorm freezing with an EWC regularization ablation experiment using **two teachers and two students**; and (iii) A privacy analysis using membership inference attack and differential privacy for mitigating the problem, along with the analysis of the privacy-utility trade-off. This is important because the clinical models need to learn from new modalities without forgetting their past learning and at the same time maintaining patient privacy.

2. Related Work

Knowledge distillation. This concept traces back to Hinton et al., who pointed out that soft outputs provided by a teacher contain more information than just a correct answer – the information about how the teacher connects classes together, and a small student can learn faster using that connection than when being taught with hard labels only. Subsequent works extended this approach by comparing not only logit outputs but intermediate representations as well. In our work, knowledge distillation plays two roles. First, it is used as an approach to compress a bulky teacher into a practical student. Second, it is used as a core of LwF, allowing the model to distill its own previous predictions.

Continual learning. The vast majority of approaches used to alleviate forgetting fit into one of three categories. Regularization approaches such as EWC and Synaptic Intelligence prevent modifications of the parameters that were important in the former task. Distillation approaches, such as LwF, maintain the output of the old model roughly constant while learning a new task. Replay approaches follow a totally different strategy and consist in storing or generating samples from the previous tasks – precisely what we are unable to do, since we want to avoid keeping old patients' scans. What usually goes unmentioned in all three types of algorithms is that of the most mundane type: normalization statistics.

Machine learning and privacy. It would be surprising for most people to learn that the trained model has memorized who it has seen. The membership inference attack relies precisely on that idea; the model will have a slightly higher confidence on the examples that it has seen during training time, which can be used to predict whether the sample has been used for training or not. This problem can be tackled using differential privacy techniques like DP-SGD, where the gradients are clipped and noise is added in order to limit the influence of each sample on the parameters of the model.

Positioning of this work. Prior related work is either focused on distillation in combination with continual learning for better retention or considers the effect of domain shift on the distribution of the normalization statistics. We diverge on three fronts. While the typical experimental protocol is to keep one teacher and one student fixed throughout, we go with the full grid between two very distinct teachers and two efficient students and thus we get the ability to isolate whether the difference between two setups is due to the actual continual learning algorithm or just the particular choice of the backbones. In addition, we opt for a particularly challenging pairing of tasks - namely X-ray vs. MRI, instead of similar datasets and thus the forgetting we observe is more of a worst-case scenario. Finally, we do not treat "no-old-data retention" as synonymous to privacy but explicitly measure the leakage and estimate its cost.

3. Method

Our continual learning experiment is based on two consecutive imaging tasks in the medical domain. In Task A, we train our model to classify chest x-rays into two classes: Normal and Pneumonia. In Task B, our task is to classify brain tumor MRIs into four categories: glioma, meningioma, no tumor, and pituitary. Our training dataset is restricted only to Task A and then we adapt our model for Task B without using any information from Task A.

$$\max_\theta \; P\_B(\theta) \quad \text{subject to} \quad P\_A(\theta) \geq \varepsilon,$$

where P_A and P_B are task-performance metrics, and ε is an acceptable retention threshold.

3.1 Proposed Workflow

This framework involves a four-step sequential pipeline (Figure 1): (1) pre-training of the teacher network on Task A; (2) knowledge transfer from teacher to a light-weight student network based on distillation of logits; (3) continual learning to adapt the student to Task B; and (4) stability control by freezing of BatchNorm. The architecture of the student network consists of multiple heads — a common backbone for feature extraction and two different heads for Tasks A and B.

3.2 Teacher and Knowledge Distillation

The teacher is a pre-trained ResNet-50 (and, in our ablation experiments, a pre-trained ViT-B/16) where the last layer is substituted with a Task-A classifier and further fine-tuned to minimize the cross-entropy loss

$$L\_CE\text{^}A = -\Sigma\_i \; y\_i\text{^}A \log f\_T(x\_i\text{^}A;\, \theta\_T).$$

Then, knowledge is distilled into the student via matching teacher's temperature scaled predictions. Specifically, given teacher logits z_T, student logits z_S, and temperature T, the loss of knowledge distillation is defined as

$$L\_KD = T^2 \cdot KL(\, softmax(z\_T/T) \,\|\, softmax(z\_S/T)\,),$$

and the student is trained using L = α·L_CE + (1−α)·L_KD, with T = 3 and α = 0.5. Temperature T determines the amount of structure in inter-class relationships revealed by the teacher: the higher the temperature means softer the output probability and thus the relative similarities that have been learned by the teacher become evident. The latter is exactly what the small student could benefit from. Thanks to the regularization effect of soft targets, the student could sometimes achieve the same or even better classification accuracy than the teacher, despite being much smaller, which happens for Task A.

3.3 Continual-Learning Strategies

EWC imposes penalties on perturbations of the parameters significant to Task A through the use of the diagonal Fisher information matrix F:

$$L\_EWC = L\_CE\text{^}B + (\lambda/2)\, \Sigma\_i\, F\_i\, (\theta\_i - \theta\_i^*)^2,$$

where θ* denotes Task-A parameters. The Learning without Forgetting (LwF) strategy enforces Task-A performance by utilizing distillation loss relative to a frozen student model, L_LwF = L_CE^B + α·L_KD^A. BN freezing is a technique where the running parameters of the BatchNorm layer (μ_BN^A, σ_BN^A) are frozen in order to avoid feature drift across domains.

3.4 Privacy Analysis

We compute privacy risk via a loss-based membership inference attack wherein a model will be more confident (less loss) on its training examples, thus, we compute a score for each example as its negative loss, and then calculate attack AUC over training member and holdout non-member examples (AUC=0.5 means no leakage). The defense involves using DP-SGD to train models where gradient per sample is computed and scaled by added noise such that influence of each patient is upper-bounded given privacy budget ε. Since DP-SGD needs per sample gradient which is not possible with BatchNorm and in-place residual connection, DP paper uses a GroupNorm CNN.

3.5 Datasets, Preprocessing and Implementation

All images are resized to 224×224 and normalized using ImageNet statistics; random horizontal flips and rotations are used in training. Task A uses the Chest X-Ray Pneumonia dataset (kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia) and Task B the Brain Tumor MRI dataset (kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset); the two datasets involve different imaging modalities, creating an enormous domain shift which causes catastrophic forgetting, making it a challenging setup for testing the proposed continual learning methods. A stratified 70/15/15 train/validation/test split is used such that class proportions are maintained in each of the splits; a subset of the training data is selected when necessary, in order to keep training computationally feasible in a single GPU. The models are built in PyTorch using Adam optimizer (learning rate 1e-3, batch size 32). Class weights are used in cross-entropy losses in order to balance the classes in Task A.

Reasoning behind the design. We use two differently structured teachers (a convolutional ResNet-50 and a transformer-based ViT-B/16) paired with two efficient students (MobileNetV3-Large and EfficientNet-B0) in order for our analysis of forgetting and the comparative ranking of continual learning approaches to be independent of a particular architecture family. The multi-head approach uses a shared backbone, the only part of the network that could potentially forget, and small heads specific for each task, allowing us to isolate forgetting and understand the effects of each continual learning strategy independently. Logit distillation is chosen over feature distillation since teachers and students in this case have very different backbones and therefore intermediate representations are not comparable, while logit outputs do not depend on any architecture specifics. Lastly, measuring retention in relation to the pre-adaptation performance of the student gives a more accurate measure of forgetting, independent of the intrinsic capacity of the model.

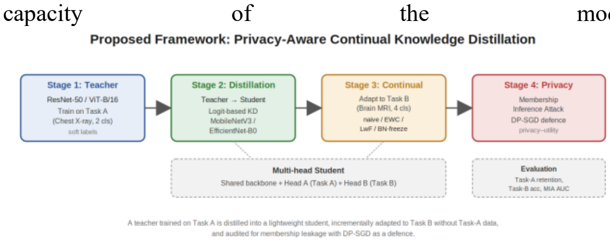

Figure 1. Proposed framework overview. A teacher trained on task A is distilled into a multi-head student that is incrementally adapted to task B without access to task A's data and verified for membership leakage using DP-SGD as a safeguard.

4. Results

First, we consider single-task performance, then we explore the impact of each of the four continual learning approaches on forgetting, perform the analysis for two teachers and two learners, weaken the strength of EWC, and finally, we consider the issue of privacy. In all these tasks, two values come into play: Task A Retention, which refers to the amount of initial skill retained during adaptation, and Task B Accuracy, which refers to the acquisition of the new task.

4.1 Teacher and Distilled Student (Task A)

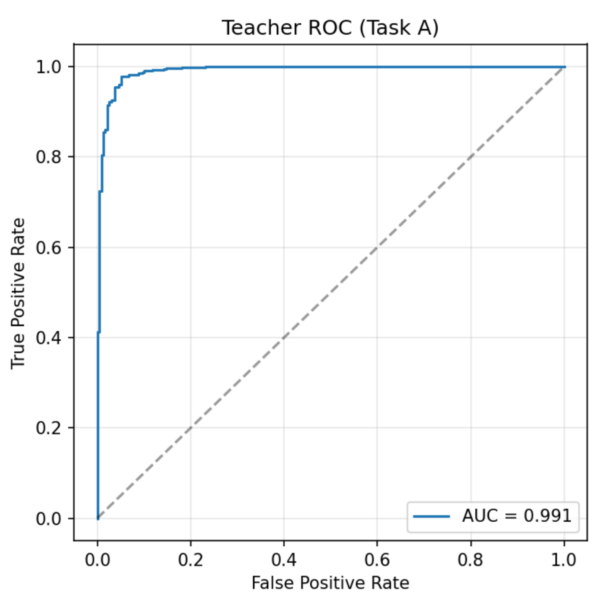

Figure 2. Teacher ROC curve on Task A (chest X-ray).

ResNet-50 Teacher achieves 95.79% on Task A Test set (AUC = 0.991). Distilled MobileNetV3 student achieves 97.04% accuracy (AUC = 0.994) - beating the teacher slightly, even though it has around five times less parameters, proving that soft target has transferred knowledge well while serving as a good regularizer.

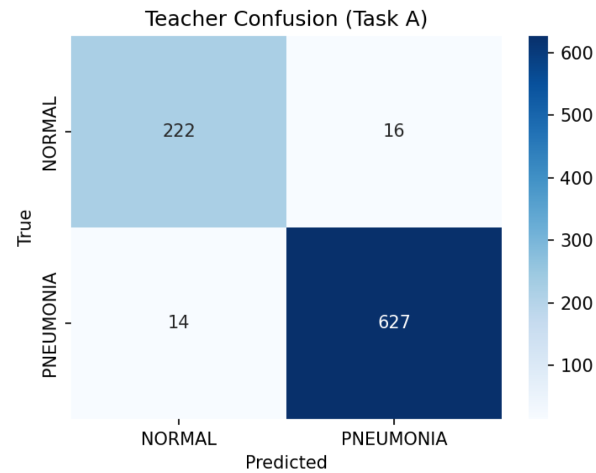

Figure 3. Teacher confusion matrix on Task A.

The confusion matrix is worth a second look. Almost all of the teacher's errors are Normal scans predicted as Pneumonia rather than the reverse, which, in a screening context, is the safer direction to fail in: a false alarm sends a patient for a second read, while a missed pneumonia does not. The student inherits this same conservative bias through the soft labels, so the behavior we care about clinically is preserved by distillation, not just the headline accuracy.

4.2 Continual Learning: Forgetting vs. New-Task Accuracy

Catastrophic forgetting occurs due to the adaptation of the distilled MobileNetV3 student network in Task B. Table 1 presents retention for Task A (in relation to pre-adaptation performance) and accuracy for Task B. LwF shows the weakest retention here, while naive fine-tuning is moderate, EWC improves retention considerably, and BatchNorm freezing reaches near-perfect retention alongside good Task B accuracy. It is easy to understand the reasoning behind such results: being performed on two different modalities, Task A and Task B lead to overwriting the BatchNorm running statistics computed on X-rays with the statistics computed on MRIs, thus corrupting the common representation of Task A even in the case when no changes are made to the head responsible for Task A.

Method	Task A ret. (%)	Task B acc.
BN-freeze	100.0	0.900
LwF	75.000	0.951
EWC	93.9	0.940
Naive (baseline)	89.6	0.927

Table 1. Continual-learning comparison (ResNet-50 → MobileNetV3). Higher retention = less forgetting.

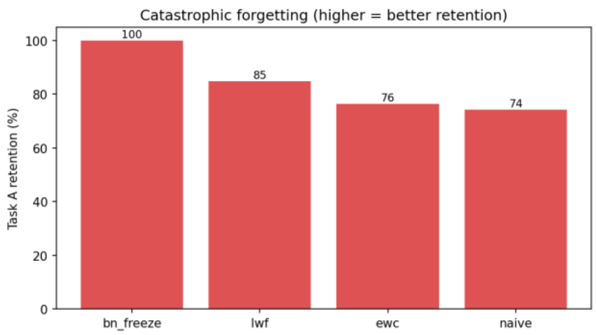

Figure 4. Task-A retention by continual-learning method.

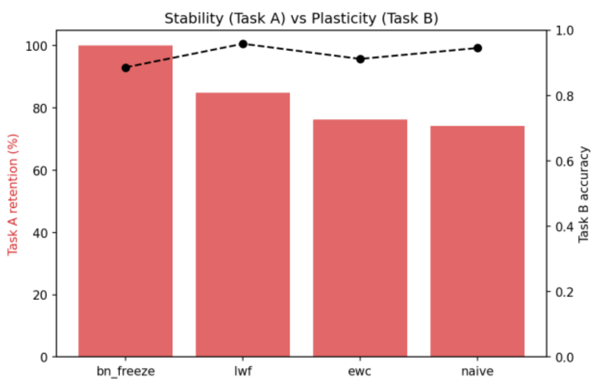

Figure 5. Stability (Task-A retention) vs. plasticity (Task-B accuracy).

4.3 Across Teachers and Students

Table 2 captures the results for all four (teacher, student) pairs. The following two trends remain constant across all four pairs: naïve fine-tuning leads to the worst performance (the Task-A retention decreases to 41.9% for ResNet-50→EfficientNet-B0 and to 27.8% for ViT→EfficientNet-B0), while BatchNorm freezing brings back ≈100% retention in all cases. The EWC method acts as a powerful regularizer for ResNet-50 backbone, and LwF proves the most effective for the ViT→EfficientNet-B0 pair.

Upon further examination, EfficientNet-B0 students catastrophically forget more when naively fine-tuned compared to MobileNetV3 students, which indicates that the process of backbone normalization and reuse of features is influential to the fragility of early knowledge. While the ViT teacher provides a representation that is more difficult to remember for small students during naive adaptation, LwF, which explicitly distills the previous outputs, succeeds in this scenario and obtains 92.1% retention. On the other hand, batch norm freezing is indifferent to all of the above factors because the frozen Task-A normalization statistics eliminate the primary failure mechanism irrespective of teacher and student, and this is why it becomes the best technique in our experiments and a reasonable default option in cross-modal continual learning.

Teacher	Student	Method	Ret.%	TaskB
ResNet-50	MobileNetV3	naive	89.6	0.927
ResNet-50	MobileNetV3	ewc	93.9	0.940
ResNet-50	MobileNetV3	lwf	75.0	0.951
ResNet-50	MobileNetV3	bn_freeze	100.0	0.900
ResNet-50	EfficientNet	naive	41.9	0.973
ResNet-50	EfficientNet	ewc	95.5	0.967
ResNet-50	EfficientNet	lwf	76.9	0.970
ResNet-50	EfficientNet	bn_freeze	100.0	0.905
ViT-B/16	MobileNetV3	naive	67.9	0.958
ViT-B/16	MobileNetV3	ewc	81.8	0.926
ViT-B/16	MobileNetV3	lwf	75.2	0.979
ViT-B/16	MobileNetV3	bn_freeze	100.0	0.869
ViT-B/16	EfficientNet	naive	27.8	0.966
ViT-B/16	EfficientNet	ewc	59.7	0.957
ViT-B/16	EfficientNet	lwf	92.1	0.966
ViT-B/16	EfficientNet	bn_freeze	100.0	0.910

Table 2. Retention and Task-B accuracy for all teacher×student×method combinations.

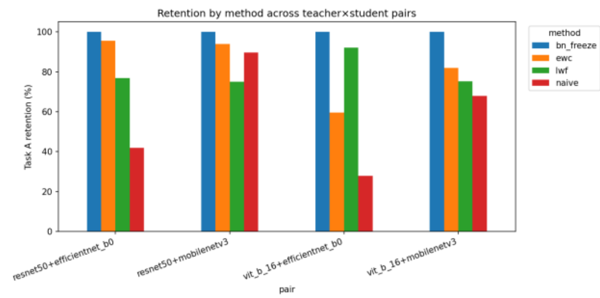

Figure 6. Task-A retention by method, grouped by teacher×student pair.

4.4 Ablation: EWC Regularization Strength

The results confirm the desired trade-off between stability and plasticity as the retention increases from 86.3% (λ = 0, i.e., the naive case) to a maximum of 95.4% (λ = 2000) before falling back to 90.6% (λ = 8000), which is the result of too much regularization, whereas the Task B accuracy remains constant.

EWC λ	Task A ret. (%)	Task B acc.
0	86.3	0.933
500	95.0	0.959
2000	95.4	0.966
8000	90.6	0.931

Table 3. EWC λ ablation.

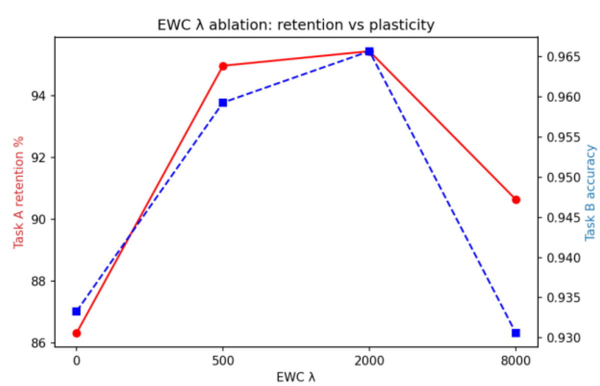

Figure 7. EWC λ ablation: retention vs. Task-B accuracy.

Although one may be tempted to interpret the graph as "bigger λ is better," the sharp fall around λ=8000 illustrates the flaw in that reasoning. At a certain point, the Fisher penalty constrains too many parameters, such that it becomes impossible for the model to shift sufficiently to learn Task B well, and both performance on Task A and learning of Task B suffer. The optimal value λ=2000 lies somewhere in between, when the Fisher penalty is strong enough to secure the relevant parameters of Task A, while allowing others to adjust.

4.5 Privacy Analysis

Table 4 shows membership inference AUC and Task-A accuracy without privacy and with DP-SGD at three different privacy budgets. Without privacy, the attack AUC is 0.526, showing that the implicit privacy from the application of distillation, augmentation, and class balancing is quite strong since the model does not memorize individual data points during training. The effect of DP-SGD on the attack AUC is to move it closer to 0.50 as ε gets smaller, as we would expect given the privacy–utility trade-off. However, this comes at the expense of lower accuracy. We thus view privacy here as a formal privacy guarantee applied to a low-leakage model.

ε (budget)	Task A acc.	MIA AUC
∞ (no DP)	0.919	0.526
8	0.854	0.512
3	0.774	0.506
1	0.861	0.503

Table 4. Privacy–utility trade-off (smaller ε = stronger privacy).

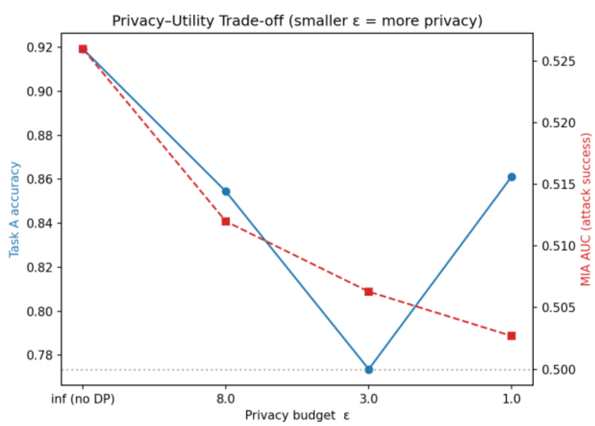

Figure 8. Privacy–utility trade-off across privacy budgets ε.

Two things should be noted here. First, that the low level of non-private leakage is an interesting result in itself, since distillation makes the decision boundary softer and augmentation prevents the model from memorizing each individual scan, making the privacy threat inherent to this pipeline minimal even without any defence whatsoever. Second, the performance of the DP models is not monotonic with regard to ε in our low-budget experiment runs – and it is reasonable, as DP-SGD adds a lot of noise to the gradients, which means that optimization gets increasingly noisy and unstable when using very small budgets, meaning that a single run with ε=1 could happen to be higher on the chart than a run with ε=3.

4.6 Computational Efficiency

Deployment is an important driver behind the need for distillation, and therefore, the comparison between the size and number of parameters of the teacher and student backbone models is provided (Table 5). The student model of MobileNetV3 is only about five times smaller than the teacher model of ResNet-50 but achieves the same or even better Task-A accuracy scores, and the same efficiency is provided by EfficientNet-B0. Thus, the model produced by the pipeline is proven to be deployable in conditions when continual learning and privacy are needed.

Model	Role	Params (M)	Rel. size
ResNet-50	Teacher	~23.5	1.0×
ViT-B/16	Teacher	~86.0	3.7×
MobileNetV3	Student	~4.2	0.18×
EfficientNet-B0	Student	~4.0	0.17×

Table 5. Backbone sizes. Students are far smaller than the teachers while retaining accuracy, supporting efficient deployment.

4.7 Confusion Analysis of the Best Configuration

The Confusion matrices for both tasks under the setting of no freezing of Batch Norm suggest that the system behavior is balanced, since the classes of Task-A are separable after the adaptation (the knowledge was retained), while Task-B classes were classified well.

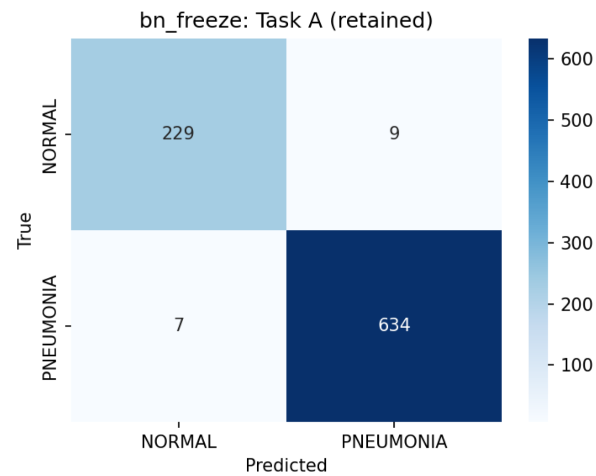

Figure 9. Confusion matrix on Task A after adaptation (retained).

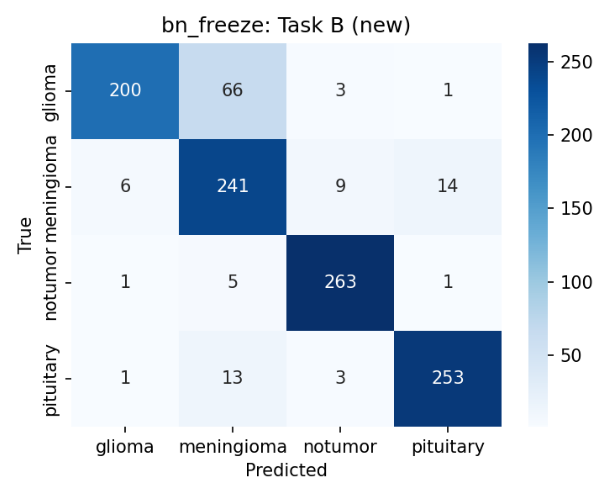

Figure 10. Confusion matrix on Task B (new task).

There are two striking aspects about this example. First, the Task-A matrix remains largely diagonal even after training the model on a wholly different modality, since the whole idea of this approach is to retain the knowledge about the X-rays. Second, on Task B, there are only a few errors, but all of them concern gliomas and meningiomas, which indeed look alike on MRI scans and even confuse radiologists, so the errors are justified. Again, no task suffers at the expense of the other – that's exactly what we were trying to achieve.

5. Conclusion

We tried to learn two entirely distinct medical imaging tasks sequentially, without forgetting what we learned from the former or revealing too much information from our training data. Distillation took care of most of the efficiency requirements: a ResNet-50 teacher was compressed down to a student of considerably smaller size with no loss of accuracy, indeed even a bit of gain. For the forgetting experiments, the outcome was rather unambiguous: with no special measures taken, naive fine-tuning obliterated Task-A performance with respect to all teachers and students we tested. Freezing the BatchNorm statistics prevented nearly all such degradation while still permitting the learning of Task B, and EWC provided a more tunable alternative. On the privacy side, the distilled and regularized models were not prone to memorizing their training data, and DP-SGD provided an extra theoretical guarantee at the price of a small loss in accuracy.

There are bounds that can be explicitly stated. In our experiment, we just considered two tasks done one after the other and a model with multiple heads; the attack considered is the easiest among the loss based attacks; differential privacy was implemented for the basic task and not the entire pipeline. The logical way to proceed is to consider a long chain of tasks, stronger attacks with the shadow models and implementing DP for the entire pipeline.

Broader impact. What is the relevance of this work to a hospital? In terms of real-world deployment of medical AI systems, it turns out to mirror ours precisely: new scanners come along, new screening needs to be done, and having to rebuild your machine learning system from scratch every time, or forgetting one of its abilities every time you learn something new, is a bad practice to adopt. Being able to incorporate a new modality while proving that your algorithm is maintaining its competency and measuring leakage is a step in the right direction towards something that a clinical environment would be comfortable using. What we are most excited about, however, is that the solution to the problem at hand, and probably the most effective one by far, is the most cost-efficient one: storing some statistics for normalization is free in both memory and computation.

Threats to validity. There are a few things to bear in mind. Firstly, privacy is tested using our experimental models running in a GroupNorm environment, as DP-SGD will not support BatchNorm nor in-place residuals of the students; this is a standard approach; however, this means the leakage numbers only apply to the stand-in model. Second, retention is calculated relative to the individual accuracy of each student prior to adaptation, which is the correct measure of forgetting, yet not the same number as absolute accuracy used elsewhere. We had to limit our training sample size, so the results would definitely improve using all the data. Finally, there was one constant thing in every test, regardless of which teacher or student we used: the ranking remained consistent - worst result for naive, best one for BatchNorm freezing method, which supports the main point of this work.

6. Author Contributions

Bishozit C. D. (Team Lead). He was responsible for managing the entire process of the project, starting from formulating the problem statement along with a privacy-preserving concept to designing the overall system architecture and writing the project proposal. He implemented the entire code base that includes a data pipeline, teacher, and multi-head students' model architecture, knowledge distillation system, four continual learning methods, a privacy module which includes membership inference attack and differential privacy SGD, a multi-pair experiment runner, and a plotting module. He ran all the experiments and generated figures and tables for our paper.

Abdullah A.N.T. Worked on the experimental pipeline and the model training process, ran and checked part of the teacher-student and continual learning experimentations locally (including the more complex ViT-B/16 experimentation) assisted in the debugging of differential privacy training process, and contributed to the Introduction and Conclusion sections.

Shaikh F.R. Worked on the review of literature and collection of relevant material, helped in preparing and organizing the data sets, helped in proofreading the manuscript, formatting the tables, and references, as well as reviewing the experimental results for the report.

Overall, all authors reviewed the final projects & agreed with the final result for submission. All authors carefully reviewed the project report carefully and agreed to submit.

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